

Stanley Decompositions and Riquier Bases

Joachim Apel*

MSRI Berkeley & University of Leipzig

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Stanley Decomposition

R – $\mathbb{N}^\nu$ -graded $\mathcal{K}$ -algebra

M – finitely generated $\mathbb{Z}^\nu$ -graded R -module

Conjecture (R.P.Stanley, 1982): M can be decomposed in a direct sum

$$M = \bigoplus_{i=1}^r f_i S_i, \quad (1)$$

where the f_i are $\mathbb{Z}^\nu$ -homogeneous elements of M and the S_i are subalgebras of R , each of them generated by finitely many algebraically independent $\mathbb{N}^\nu$ -homogeneous elements and satisfying $\dim_{\mathcal{K}} S_i \geq \text{depth} M$.

Special case: $R = \mathcal{K}[x_1, \dots, x_n]$ with natural $\mathbb{N}^n$ -grading, that is, $\mathbb{N}^n$ -homogeneous means monomial. Then the decomposition looks like

$$M = \bigoplus_{i=1}^r f_i \mathcal{K}[x_{j_1}, \dots, x_{j_{m_i}}],$$

where $m_i \geq \text{depth} M$ for all $i \in \{1, \dots, r\}$.

Depth of a graded module over an $\mathbb{N}$ -graded ring

A sequence $p_1, \dots, p_d$ of homogeneous elements of $\langle X \rangle$ is called *regular* for M , if p_1 is not a zero-divisor in M and p_{i+1} is not a zerodivisor in $M/(p_1, \dots, p_i)M$ for all $1 \leq i < d$. The maximal length of a regular sequence for M is called the *depth* of M and denoted by $\text{depth}M$.

Note, though, we consider graded modules over $\mathbb{N}^\nu$ -graded rings, the notion of depth refers **always** to a specialization to an $\mathbb{N}$ -grading.

Special case $\nu = 1$

- For Cohen-Macaulay modules M it is well-known that M is a free module over an arbitrary subalgebra of R generated by a parameter system of M .
- For $\mathcal{K}$ infinite a method of Rees (1956) provides a Stanley decomposition, a general method is due to Baclawski and Garsia (1981). Both methods are not full algorithmic.
- Full algorithmic solution due to Sturmfels and White (1991)
- In the literature the notion Stanley decomposition sometimes refers only to the special case $\nu = 1$.

Riquier Bases (Involution Bases)

$R = \mathcal{K}[X]$ – a polynomial ring over the field $\mathcal{K}$ in the indeterminates $X = \{x_1, \dots, x_n\}$

$M \subseteq R^k$ – submodule of a free R -module of rank k

$\prec$ – admissible term order of R^k

A finite set $V \subset M$ is called a (minimal) *Riquier basis* of M w.r.t. $\prec$ if there exists a linear order $\sqsubset$ of $\text{It}_\prec(V)$ and a family $(Y_v)_{v \in V}$ of subsets of X such that

$$\text{in}_\prec(M) = \bigoplus_{v \in V} \text{in}_\prec(v) \mathcal{K}[Y_v]$$

and for all $v, v' \in V$ such that $\text{It}_\prec(v') \sqsubset \text{It}_\prec(v)$

$$\text{in}_\prec(v) \mathcal{K}[Y_v] \cap \text{in}_\prec(v') \mathcal{K}[X] = \{0\} .$$

Minimal Riquier bases correspond to passive complete orthonomic systems of linear PDEs with constant coefficients.

Riquier bases & Stanley decomposition

$R = \mathcal{K}[x_1, \dots, x_n] - \mathbb{N}^n$ graded in the usual way

$R^k - \mathbb{Z}^n$ -graded extending the $\mathbb{N}^n$ -grading of R

M – graded submodule of R^k

Def. 1 (A., 1998) *A minimal Riquier basis $V = \{v_1, \dots, v_r\}$, all v_i homogeneous, of M with multiplier family $(Y_i)_{i \in \{1, \dots, r\}}$ yields a decomposition*

$$M = \bigoplus_{i=1}^r v_i \mathcal{K}[Y_i] .$$

This is a Stanley decomposition in case

$$\text{depth} M \leq \min |Y_i| .$$

- Special case $k = 1$ and M in generic coordinates: Pommaret bases provide Stanley decompositions (Bayer and Stillmann 1987, Mall 1998)

Maximal multiplier families

A family $(Y_i)_{i \in \{1, \dots, r\}}$ of multipliers is called *admissible* for the ordered set $V = \{v_1, \dots, v_r\}$, $v_1 \sqsubset v_2 \sqsubset \dots \sqsubset v_r$, of monomial elements of R^k if for any $1 \leq i < j \leq r$ it holds either

$$\text{It}_{\prec}(v_i)\mathcal{K}[X] \cap \text{It}_{\prec}(v_j)\mathcal{K}[Y_j] = \{0\}$$

or

$$\text{It}_{\prec}(v_i)\mathcal{K}[Y_i] \subset \text{It}_{\prec}(v_j)\mathcal{K}[Y_j]$$

If in addition, the admissibility of $(Z_i)_{i \in \{1, \dots, r\}}$, where $Y_i \subseteq Z_i$, requires equality at all places, we call $(Y_i)_{i \in \{1, \dots, r\}}$ a *maximal* multiplier family for the ordered set V .

All classical multiplier settings (Janet 1929, Pommaret 1978, Thomas 1937) are admissible for arbitrary sets of monomials for $\sqsubset$ being a reverse lexicographical order.

But none of them yields maximal multiplier families for all finite sets of monomials (A. 1998).

Riquier and Stanley dimension

For a Riquier basis $V = \{v_1, \dots, v_r\}$ of M with multiplier family $(Y_i)_{i \in \{1, \dots, r\}}$ we set

$$\text{Rdim } V := \min_{1 \leq i \leq r} |Y_i|.$$

Furthermore, we define:

$$\text{Rdim } M := \max_V \text{Rdim } V \quad \text{and}$$

$$\text{Sdim } M := \max_{(1)} \min_{1 \leq i \leq r} |S_i|$$

Obviously, for any graded module M we have:

$$\text{Rdim } M \leq \text{Sdim } M \tag{2}$$

There are modules M for which Inequality (2) is proper. (see Examples 2 and 3)

In what follows, we will always assume that the considered Riquier bases are **minimal** and the corresponding multiplier families are **maximal**.

Monomial ideals M , i.e. special case $k = 1$

Crit. 1 (A., 2001) Assume M possesses an (ordered) Riquier basis $V = \{v_1, \dots, v_r\}$ (with multiplier family $(Y_j)_{j \in \{1, \dots, r\}}$) such that for some $1 \leq i \leq r$ and some $x_m \in Y_i$ we have:

1. $|Y_i| \leq |Y_l|$ for all $l \neq i$,
2. for each $x_j \notin Y_i$, $1 \leq j \leq n$, there exists $k_j \in \{1, \dots, i-1\}$ such that
 - (a) $\deg_{x_j} v_i = \deg_{x_j} v_{k_j} - 1$,
 - (b) $\deg_{x_l} v_i \geq \deg_{x_l} v_{k_j}$ for all $l \neq j$ such that $x_l \notin Y_i$,
 - (c) $\deg_{x_m} v_i > \deg_{x_m} v_{k_j}$.

Then $\text{depth} M \leq |Y_i| \leq \text{Rdim } M \leq \text{Sdim } M$, i.e. Stanley's Conjecture holds.

Example 1 $M = \langle xyz, x^2, y^2, z^2 \rangle \subset \mathcal{K}[x, y, z]$

(ordered) Riquier basis:

$$\{z^2, y^2z, x^2yz, x^2z, y^2, x^2y, x^2, xyz\}$$

Corresponding multipliers: $Y_{z^2} = \{x, y, z\}$, $Y_{y^2z} = Y_{y^2} = \{x, y\}$, $Y_{x^2yz} = Y_{x^2z} = Y_{x^2y} = Y_{x^2} = \{x\}$, $Y_{xyz} = \emptyset$

Criterion fails!

(ordered) Riquier basis:

$$\{z^2, y^2z, \underline{xyz}, x^2z, \underline{y^2}, \underline{\underline{x^2y}}, x^2\}$$

Corresponding multipliers: $Y_{z^2} = \{x, y, z\}$, $Y_{y^2z} = Y_{y^2} = \{x, y\}$, $Y_{xyz} = Y_{x^2z} = Y_{x^2y} = Y_{x^2} = \{x\}$

Criterion holds! That is, $\text{depth}M \leq 1$, which must be equality.

Example 2 $M = \langle yu, xu, yz, xz \rangle \subset \mathcal{K}[x, y, z, u]$

That is already an (ordered) Riquier basis with corresponding multipliers $Y_{yu} = \{x, y, z, u\}$, $Y_{xu} = \{x, z, u\}$, $Y_{yz} = \{x, y, z\}$, $Y_{xz} = \{x, z\}$

Criterion fails for this and any other Riquier basis!

Nevertheless, $\text{depth} M = 2 = \text{Rdim } M$.

But, $\text{Sdim } M = 3$ since $M = xyzu\mathcal{K}[x, y, z, u] \oplus yu\mathcal{K}[x, y, u] \oplus xu\mathcal{K}[x, z, u] \oplus yz\mathcal{K}[y, z, u] \oplus xz\mathcal{K}[x, y, z]$.

Example 3

$$M = \langle yvw, xvw, zuw, xuw, yzw, zuv, yuv, xzv, xyu, xyz \rangle \subset \mathcal{K}[x, y, z, u, v, w]$$

$xyz \{x, y, z\}$, $xyu \{x, y, z, u\}$, $xzv \{x, y, z, v\}$, $yuv \{x, y, u, v\}$, $zuv \{x, y, z, u, v\}$, $yzw \{x, y, z, w\}$, $xuw \{x, y, u, w\}$, $zuw \{x, y, z, u, w\}$, $xvw \{x, z, v, w\}$, $yvw \{x, y, z, v, w\}$, $xuvw \{x, u, v, w\}$, $yuvw \{x, y, u, v, w\}$, $zuvw \{x, y, z, u, v, w\}$ is a best possible Riquier basis (including the corresponding multiplier sets) and, again, the criterion fails for any Riquier basis!

But now, $\text{depth} M = 4 > \text{Rdim } M = 3$.

Nevertheless, $\text{Sdim } M = 4$ since

$$\begin{aligned} M = & xyz\mathcal{K}[x, y, z, w] \oplus xyu\mathcal{K}[x, y, z, u] \oplus xzv\mathcal{K}[x, y, z, v] \oplus yuv\mathcal{K}[x, y, u, v] \oplus zuv\mathcal{K}[y, z, u, v] \oplus yzw\mathcal{K}[y, z, v, w] \\ & \oplus xuw\mathcal{K}[x, y, u, w] \oplus zuw\mathcal{K}[x, z, u, w] \oplus xvw\mathcal{K}[x, z, v, w] \oplus yvw\mathcal{K}[x, y, v, w] \oplus xuvw\mathcal{K}[x, u, v, w] \oplus yuvw\mathcal{K}[y, u, v, w] \\ & \oplus zuvw\mathcal{K}[z, u, v, w] \oplus yzuw\mathcal{K}[y, z, u, w] \oplus xzuv\mathcal{K}[x, z, u, v] \oplus xzuw\mathcal{K}[x, z, u, w] \oplus xyuvw\mathcal{K}[x, y, u, v, w] \\ & \oplus yzuvw\mathcal{K}[x, y, z, u, v, w]. \end{aligned}$$

Example 4 $M = \{xz, yz, zu, u^2\} \subset \mathcal{K}[x, y, z, u]$

Janet basis, minimal in the sense of Gerdt and Blinkov (1998): $xz \{x, y, z, u\}, xu^2 \{x, y, u\}, yz \{y, z, u\}, yu^2 \{y, u\}, zu \{z, u\}, u^2 \{u\}$

But Criterion applies to the Riquier basis:

*$\underline{xz} \{x, y, z, u\}, \underline{yz} \{y, z, u\}, \underline{zu} \{z, u\}, u^2 \{x, y, u\}$
confirming $\text{depth} M \leq 2$, which in fact is equality.*

General results

Classes of monomial ideals for which the existence of Riquier basis satisfying our Criterion, and hence the validity of Stanley's Conjecture, could be proved (A. 2001):

1. monomial ideals M such that $\text{depth} M \leq 2$,
2. all monomial ideals M in case $n \leq 3$,
3. M generated by a set of monomials such that for some fixed variable x_i , whenever $\deg_{x_i} m = \deg_{x_i} m'$ for two distinct generators then there is a third generator m'' such that $m'' \mid \text{lcm}(m, m')$ and $\deg_{x_i} m'' < \deg_{x_i} \text{lcm}(m, m')$.

Includes all strongly generic monomial ideals (Bayer, Peeva and Sturmfels 1998) and all generic monomial ideals (Miller, Sturmfels and Yanagawa 2000).

Special case $M = R/I$

Also strongly connected to Riquier's Existence Theorem, but *not* to algebraic Riquier bases.

If $I \subset R$ is related to an (orthonomic) system of PDEs then Riquier's Existence Theorem requires not only the decomposition of I (principal derivatives) but also that of R/I (parametric derivatives). The latter decomposition provides the initial determinations for solving the system. We always have:

$$\text{Sdim } M \leq \min_{\mathfrak{p} \in \text{Ass}(M)} \text{codim } \mathfrak{p} \quad (3)$$

$$\text{depth } M \leq \min_{\mathfrak{p} \in \text{Ass}(M)} \text{codim } \mathfrak{p} \quad (4)$$

Hence, Stanley's Conjecture holds whenever we have equality in (3). In the most important special case of Cohen-Macaulay modules M this relation is even *iff* since then we have equality in (4).

Case $\text{Sdim } M = \min_{\mathfrak{p} \in \text{Ass}(M)} \text{codim } \mathfrak{p}$

Example 5 *Strongly generic ideal* $I \subset \mathcal{K}[x, y, z, w]$

$$I = (x^2, y^2 z^2, yzw, w^2) = (x^2, y, w^2) \cap (x^2, z, w^2) \cap (x^2, y^2, w) \cap (x^2, z^2, w)$$

$$M = R/I = \mathcal{K}[z] \oplus y \mathcal{K}[y] \oplus yz \mathcal{K}[z] \oplus y^2 z \mathcal{K}[y] \oplus x \mathcal{K}[z] \oplus w \mathcal{K}[z] \oplus xw \mathcal{K}[z] \oplus yw \mathcal{K}[y] \oplus xy \mathcal{K}[y] \oplus xyw \mathcal{K}[y] \oplus xyz \mathcal{K}[z] \oplus xy^2 z \mathcal{K}[y]$$

Large example classes (A. 2001):

1. $n \leq 3$,
2. Generic monomial ideals I ,
3. Cogeneric monomial Cohen-macaulay ideals I , as defined in (Miller, Sturmfels and Yanagawa 2000)

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